

Chapter 9: Database

A database is a collection of organized data that allows for efficient storage, retrieval, and management of large volumes of information. This chapter transitions from manual systems (like phone books) to electronic Database Management Systems (DBMS).

9.1 Basic Database Components

- **Table:** The basic unit of a database where data is stored in rows and columns.
- **Field (Column):** A specific category of information (e.g., "Name", "Phone Number").
- **Record (Row):** A complete set of data for a single entity (e.g., all details for one student).

9.2 Key Characteristics & Advantages

- **Data Redundancy:** Avoiding the storage of the same data in multiple places.
- **Consistency:** Ensuring data remains accurate across the system.
- **Data Independence:** Separating data from the application programs that use it.
- **Security:** Protecting information through access controls.

9.3 Keys in a Database

- **Primary Key:** A field (or group of fields) that uniquely identifies each record in a table (e.g., Admission Number). It cannot be null.
- **Foreign Key:** A field in one table that links to the primary key of another table, creating a relationship between them.

9.4 Relationships

Relational databases connect tables using common fields. Types include:

- **One-to-One:** Each record in Table A is linked to only one record in Table B.
- **One-to-Many:** A record in Table A is linked to multiple records in Table B (e.g., one teacher to many students).
- **Many-to-Many:** Multiple records in Table A are linked to multiple records in Table B.

9.5 Database Objects

Object	Purpose
Forms	Used for user-friendly data entry and viewing.
Queries	Used to search, filter, and retrieve specific information.
Reports	Used to present data in a structured, printable format.

Summary

- Electronic databases provide superior speed, accuracy, and sharing capabilities compared to manual systems.
- Relational databases use keys to maintain integrity and link data efficiently.
- DBMS tools like Microsoft Access or LibreOffice Base provide the interface to manage these objects.